# GRPO Explained

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# Why Do We Need RL?

Why is SFT not enough? Why do we put RL on top?

- SFT teaches the LLM to reproduce the behavior that's currently in the dataset.
- With RL, the model is encouraged to discover.

Example: User asks, "What's the weather tomorrow?"

SFT is like watching a driving instructor. RL is like getting behind the wheel, driving and learning.

Why do we need (usually) SFT before RL?

No SFT → 1000 attempts → 0 correct

SFT → 1000 attempts → 150 correct

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# What Problem Does GRPO Solve?

RLHF: Compare the rollout against a learned expectation from the critic model.

GRPO: Compare the rollout against the average of the rollouts generated for the same prompt (which becomes a group).

The critic model is expensive. With GRPO, we don't need it.

We can think of GRPO and RLHF as a student taking an exam.

# GRPO Loss

$$L(\theta) = \frac{1}{N}\sum_{i=1}^{G}\sum_{t=1}^{T_i}[-S_{i,t}(\theta) + \beta D_{i,t}(\theta)]$$

Where

- $G$ is the number of groups
- $T_i$ is the $i$'th generated rollout within the group $G$.
- $t$ is the $t$'th token within $T_i$
- $N$ is the total number of valid generated tokens within the current group.

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# GRPO Loss (logprobs)

$$\ell_{i,t}^{\theta} = \ln \pi_{\theta}(y_{i,t} \mid x, y_{i,<t})$$ (new_logprobs)

$$\ell_{i,t}^{\text{old}} = \ln \pi_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})$$ (old_logprobs)

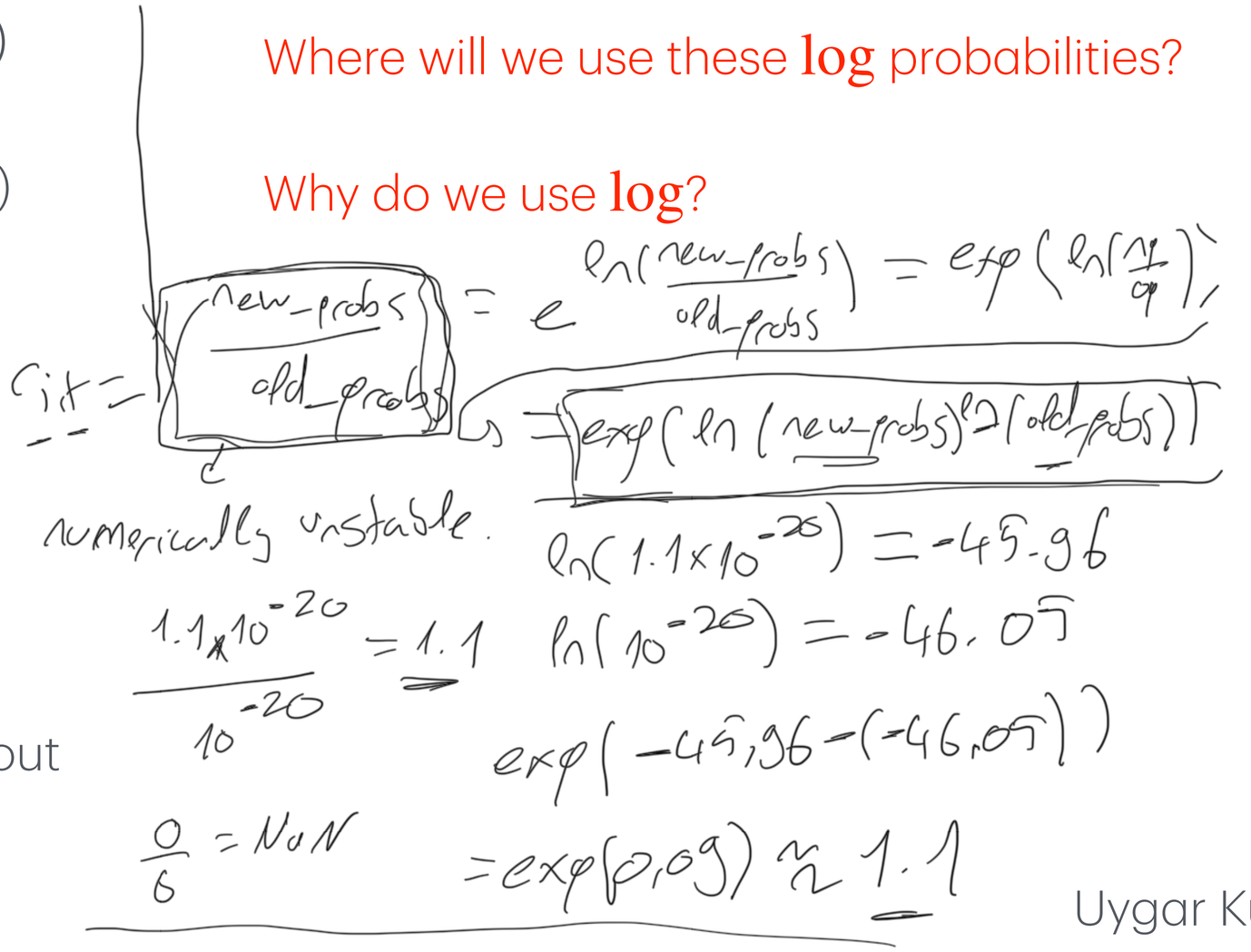
$$\ell_{i,t}^{\text{ref}} = \ln \pi_{\text{ref}}(y_{i,t} \mid x, y_{i,<t})$$ (ref_logprobs)

Where

- $i$ is the index of the current rollout
- $t$ is the current token in the current rollout
- $x$ is the prompt

Where will we use these **log** probabilities?

Why do we use **log**?

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# GRPO Loss (Surrogate Function)

Since we're trying to minimize $L(\theta)$, we have to maximize clipped surrogate $S_{i,t}$.

$$S_{i,t}(\theta) = \min[r_{i,t}(\theta)\hat{A}_i,\ \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_i]$$

Where the probability ratio $r_{i,t}$ is $r_{i,t}(\theta) = \dfrac{\pi_\theta\left(y_{i,t} \mid x, y_{i,<t}\right)}{\pi_{\theta_{\text{old}}}\left(y_{i,t} \mid x, y_{i,<t}\right)}$, which equals, and is calculated with

$r_{i,t}(\theta) = \exp\left(\ell_{i,t}^{\theta} - \ell_{i,t}^{\text{old}}\right)$ while implementing for numerical stability.

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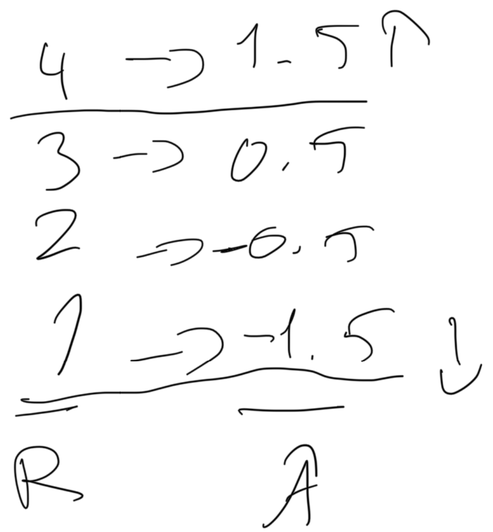
# GRPO Loss (Surrogate Function)

And where advantage $\hat{A}_i$ is $\hat{A}_i = \dfrac{R_i - \overline{R}}{\sigma_R + \delta}$

Where

- $R_i$ is the reward of rollout $i$
- $\overline{R}$ if the average reward of the group
- $\sigma_R$ is the standard deviation of the group rewards (we divide by it to normalize the scale of the advantages)
- $\delta$ is a small constant to prevent division by 0

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# GRPO Loss (Surrogate Function)

The advantage tells us how well this rollout performed compared with the other rollouts in the same group.

- A positive advantage $\hat{A}$ reinforces that rollout behavior
- A negative advantage $\hat{A}$ discourages that rollout behavior

The larger the magnitude of the advantage, the stronger the learning signal.

Why $\hat{A}$ behaves this way? Let's put things together (next slide).

# GRPO Loss (Surrogate Function)

$$S_{i,t}(\theta) = \min[r_{i,t}(\theta)\hat{A}_i, \text{clip}(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_i]$$

$$r_{i,t}(\theta) = \frac{\pi_\theta\left(y_{i,t} \mid x, y_{i,<t}\right)}{\pi_{\theta_{\text{old}}}\left(y_{i,t} \mid x, y_{i,<t}\right)}$$

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# GRPO Loss (KL Penalty)

Regular KL divergence would require us to go over the full vocabulary with:

$D_{\mathrm{KL}}(\pi_\theta \| \pi_{\mathrm{ref}}) = \sum_{v \in \mathcal{V}} \pi_\theta(v \mid s) \log \frac{\pi_\theta(v \mid s)}{\pi_{\mathrm{ref}}(v \mid s)}$. This requires a lot of calculations. However, we have the sampling $y \sim \pi_\theta(\cdot \mid s)$. So, instead, we can do a much cheaper estimate.

For GRPO, Schulman's k3 KL estimator is used. Which is the following:

$$D_{i,t}(\theta) = \exp\left(\ell_{i,t}^{\mathrm{ref}} - \ell_{i,t}^{\theta}\right) - \left(\ell_{i,t}^{\mathrm{ref}} - \ell_{i,t}^{\theta}\right) - 1$$

This value is always positive.

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# GRPO Loss (Let's Put Things Together)

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{G} \sum_{t=1}^{T_i} [-S_{i,t}(\theta) + \beta D_{i,t}(\theta)]$$

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# GRPO Algorithm (Pseudocode)

```
reference_policy = frozen_copy(policy)

for prompt in dataset:
    rollouts = generate_rollouts(policy, prompt, group_size)
    old_logprobs = logprobs(policy, prompt, rollouts)
    reference_logprobs = logprobs(reference_policy, prompt, rollouts)
    rewards = reward(rollouts)
    advantages = normalize_within_group(rewards)

    for _ in ppo_epochs:
        new_logprobs = logprobs(policy, prompt, rollouts)
        ratio = exp(new_logprobs - old_logprobs)
        surrogate = min(ratio · advantages, clip(ratio, 1 - clip_epsilon, 1 + clip_epsilon) · advantages)
        kl = exp(reference_logprobs - new_logprobs) - (reference_logprobs - new_logprobs) - 1
        loss = mean(-surrogate + kl_coefficient · kl)
        update_policy(loss)
```

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